Covalent Organic Frameworks for Photocatalytic Hydrogen Evolution: Linkage Chemistry, Band-Structure Engineering, and the Path to Scalable Solar Fuels

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Abstract

Covalent organic frameworks (COFs) have moved from a crystallographic curiosity to one of the most designable classes of organic photocatalysts for the hydrogen evolution reaction (HER). Yet reported activities span four orders of magnitude, and it is rarely obvious which of the many co-varying structural choices deserves the credit. This Review organizes the evidence along a single causal chain – linkage chemistry sets conjugation and chemical stability; conjugation and stacking set the band structure; band edges and exciton binding govern charge separation; and interfacial engineering with cocatalysts converts separated charges into hydrogen – rather than by chronology. We first map the four dominant linkage families (β-ketoenamine, imine, vinylene/sp^2-carbon, and triazine) onto this chain, then survey the design levers that act on each link: donor–acceptor backbone engineering, linkage protonation, single-atom and low-loading cocatalysts, and Z/S-scheme heterojunctions. We treat mechanistic spectroscopy as an inference chain and collect matched-protocol performance comparisons with explicit irradiation, donor, and normalization variables, including apparent quantum yields up to ~80% at 420 nm and rates approaching 400 mmol g^{-1} h^{-1} in the best-reported systems. As a falsifiable synthesis of the design rules, we outline a pre-registered improvement route for the workhorse β-ketoenamine COF TpPa-1, with quantitative gates covering structure retention, mechanistic gain, activity, and durability. We close with the measurement, stability, and reactor-scale questions that separate laboratory quantum efficiency from scalable solar fuel production.

1 Introduction

Solar hydrogen is attractive because it stores intermittent sunlight in a dispatchable chemical bond, yet particulate photocatalysis remains limited by a chain of losses – photon harvesting, exciton dissociation, charge extraction, and surface turnover – each of which can silently consume an order of magnitude in efficiency [28, 62, 94]. Early oxide studies established that carrier lifetime, not absorption, is often the controlling variable: hole lifetimes in TiO_2 correlate directly with water-splitting ability, and oxygen quantum yields depend strongly on light intensity [94]. Any candidate material class therefore has to be judged not on a single headline rate but on how well its structure can be engineered along every link of that chain [12, 26].

Covalent organic frameworks (COFs) enter this competition with an unusual proposition: fully organic, crystalline, porous semiconductors whose band structure, exciton physics, and active-site placement are all encoded in molecular building blocks [60, 68, 80, 87]. The reticular toolbox unites the tunability of molecular photochemistry with the robustness and long-range order of extended solids [3, 30, 49], and two-dimensional (2D) COFs in particular stack conjugated sheets into columnar π-arrays that support exciton migration and out-of-plane charge transport [47, 96, 140]. Since the first demonstration that a hydrazone-linked COF evolves hydrogen under visible light, the field has expanded through azine, imine, β-ketoenamine, triazine, and vinylene linkage generations [32, 77, 86, 112], with recent systems reporting apparent quantum efficiencies (AQE) above 80% at 450 nm [51] and mass-normalized rates approaching 400 mmol g^{-1} h^{-1} [23,

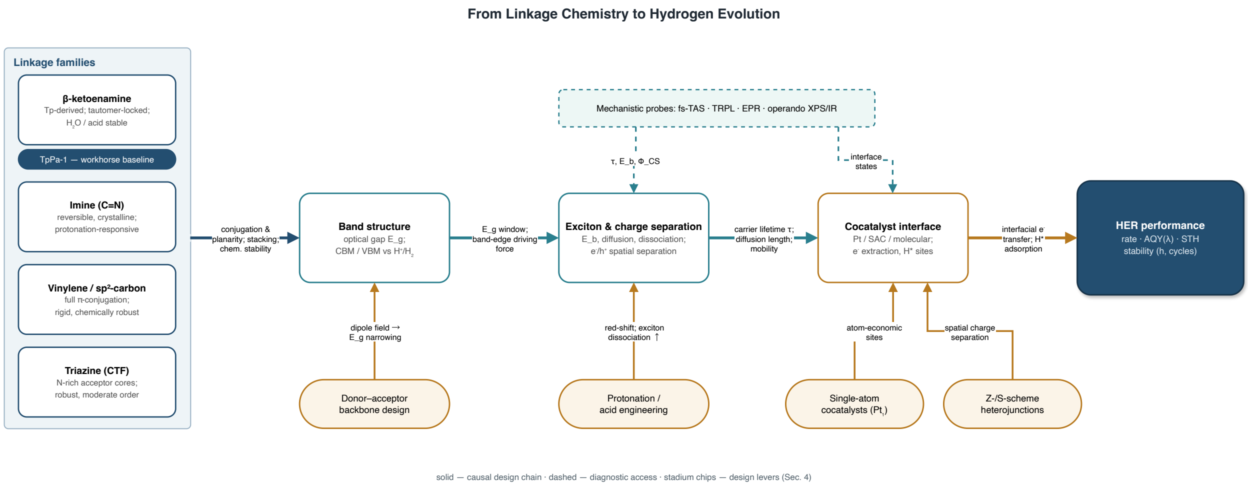

Figure 1: The causal design chain that organizes this Review. Linkage chemistry (left) sets conjugation, planarity, and chemical stability; these determine the band structure; band edges and exciton binding govern charge separation; and the cocatalyst interface converts extracted electrons into hydrogen. Solid arrows denote the causal chain, stadium chips denote the design levers of Section 2, and the dashed channel marks the mechanistic probes of Section 3.

136]. Designability, however, does not by itself establish efficient charge separation or scalable operation: strongly bound Frenkel excitons, layer disorder, and cocatalyst dependence remain recurring bottlenecks [4, 85, 128].

The literature now contains hundreds of COF photocatalysts but comparatively few controlled comparisons, and reviews that organize the material by publication date tend to obscure which structural lever actually moved which observable [1, 116, 120, 142]. This Review therefore organizes evidence by the causal design chain summarized in Figure 1: linkage chemistry → conjugation and planarity → band gap and band-edge positions → exciton dissociation and charge separation → cocatalyst interface → HER performance. Each design lever – donor–acceptor (D–A) backbone engineering, linkage protonation, single-atom cocatalysts, and Z/S-scheme heterojunctions – is treated as an intervention on a specific link, and mechanistic spectroscopy is treated as the measurement that can distinguish competing explanations [12, 26, 89].

The remainder of the Review follows this chain. Section 2 maps the linkage families and the design levers acting on band structure and exciton physics. Section 3 reviews the mechanistic characterization toolbox, from femtosecond transient absorption to operando spectroscopy. Section 4 examines synthesis methodology and crystallinity control, the practical variables that gate every other claim. Section 5 compares reported performance under explicit protocol variables. Section 6 converts the accumulated design rules into a pre-registered, falsifiable improvement route for the workhorse β-ketoenamine COF TpPa-1. Section 7 confronts the stability, measurement, and reactor questions that stand between laboratory AQE and solar fuel production, and Section 8 concludes.

2 From Linkage Chemistry to Band Structure: The Design Levers

2.1 Four linkage families, four positions on the stability–conjugation frontier

The linkage is the single decision that most strongly constrains everything downstream, because it fixes both the degree of π-delocalization across the framework and the hydrolytic stability that photocatalysis in water demands [30, 67, 80, 87]; linkage-first taxonomies now organize COF photocatalysis across reactions well beyond HER [115]. Table 1 summarizes the four families that dominate HER work.

β-Ketoenamine (Tp-based) COFs, formed from 1,3,5-triformylphloroglucinol (Tp) and diamines

Table 1: The four dominant linkage families for COF photocatalytic HER. Rates depend strongly on protocol (Section 5); exemplars are indicative, not rankings.

Family	Formation	Conjugation / planarity	Stability	HER exemplar
β-Ketoenamine	Tp + diamine; tautomer-locked	Interrupted in-plane π; ESIPT-active	Boiling water, strong acid	TpPa-1 and Tp-COF composites [97, 112]
Imine (C=N)	Reversible Schiff base	Moderate; protonation-responsive	Good; acid-sensitive linkage	Protonated D–A imine COFs, 20.7 mmol g^{-1} h^{-1} [114]
Vinylene / sp^2c	Aldol / Knoevenagel / HWE	Full π-conjugation; rigid	9 M HCl / 9 M NaOH	DBC-sp^2c-COF, 105 mmol g^{-1} h^{-1} (saltwater) [8, 122]
Triazine (CTF-like)	Nitrile cyclization; polycondensation	Planar N-rich acceptor cores	Robust; moderate order	TP-COF / CTF heterostructures [33, 110]

through combined Schiff-base condensation and irreversible enol-to-keto tautomerization, trade some in-plane conjugation for exceptional acid and water stability, which is why this family – with TpPa-1 as its workhorse – accounts for the largest share of reported COF photocatalysts and composites [92, 97, 112]. The keto-locked backbone also supports excited-state intramolecular proton transfer (ESIPT), whose induced photoisomerization must be considered when assigning activity differences within the family [105]. Imine (C=N) COFs retain reversible linkage formation and therefore high crystallinity, and their lone-pair-rich linkages are protonation-responsive, a property Section 2.3 develops into a full design lever [5, 59, 114]. Vinylene and sp^2-carbon frameworks extend conjugation through C=C bonds that cannot hydrolyze, combining full π-delocalization with chemical robustness: sp^2-carbon COFs manage the complete photon-to-charge event sequence within one lattice [47], survive 9 M HCl and 9 M NaOH [8], and, in a matched comparison, an sp^2-carbon-linked COF catalyzed overall water splitting with an AQE of 2.53% at 420 nm where its imine-linked analogue was inactive [11]. Kagome-lattice sp^2c-COFs templated from an imine precursor reach 105.3 and 80.0 mmol g^{-1} h^{-1} in 3 wt% saltwater and natural seawater, respectively [122], and pyrene-based sp^2c frameworks outperform their imine counterparts for the same topology [34, 58]. Extending the acetylenic logic, a graphdiyne-like diacetylene-rich linker delivers 10.16 mmol g^{-1} h^{-1} against 3.71 and 1.13 for its non-GDY analogues [53]. Aldol, Knoevenagel, and Horner–Wadsworth–Emmons condensations now give synthetic access across this family [77]. Finally, triazine-based frameworks contribute nitrogen-rich acceptor cores with planar, fully conjugated attributes shared with g-C_3N_4 and CTFs [33]; base- and temperature-controlled routes now deliver crystalline triazine-sp^2c hybrids [125], and the natural affinity of triazine cores for g-C_3N_4 interfaces carries over to junction chemistry beyond HER [78].

Within each family, linkage-level editing tunes activity further: cyano versus unsubstituted vinylene isomers show distinct structure–activity relationships [70], single-atom skeletal editing opens irreversible Katritzky-type routes to acridinium frameworks [76], and a matched three-linkage platform (imine, enamine, and alkene on the same knot–linker pair) isolates the linkage contribution directly, with the alkene member reaching an AQE of 6.7% at 420 nm [69]. The emerging design rule is that conjugation and hydrolytic robustness need not be traded off – vinylene chemistry achieves both – but crystallinity cost and synthetic accessibility still favor imine and β-ketoenamine platforms for systematic studies [77, 106].

2.2 Donor–acceptor backbone engineering

The controlled variable in donor–acceptor (D–A) design is the intramolecular dipole field, and its expected consequence is twofold: a narrowed optical gap and a driving force that pre-separates the Frenkel exciton along the donor-to-acceptor axis [55, 56, 89]. Computation now leads synthesis here: density-functional screening of triazine- and benzene-cored acceptor series shows exciton binding energy (E_b) varies systematically with donor strength, and the synthesized activity order follows the calculated E_b ranking [81]. The benchmark experimental results are consistent: a β-ketene–cyano D–A pair prolongs carrier lifetime dramatically and delivers an AQE of 82.6% at 450 nm with Pt [51]; a pyrene-donor/dual-benzothiadiazole-acceptor COF averages 12.7 mmol g^{-1} h^{-1} under visible light, among the highest for D–A-type systems [55]; and trithiophene-based D–A frameworks operate cocatalyst-free [71].

Three refinements sharpen the basic rule. First, symmetry matters as much as strength: highly symmetric D–A architectures cancel their dipoles and weaken the built-in electric field, so breaking symmetry – excitonic dipole reorientation [4], asymmetric imine-linker orientation (raising solar-to-hydrogen efficiency from 2.76% to 4.24% [59]), or ground-state charge-transfer bonding that directly intensifies the built-in field [91] – recovers the driving force. Multistep charge transfer in single-crystalline COFs shows how far directional field design can be pushed [48]. Second, the linker length and core conjugation co-tune the gap: biphenyl linkers outperform benzene and triphenyl analogues in a matched isoreticular series [17], progressive core π-extension promotes exciton dissociation [52], benzobisoxazole-bridged fully conjugated frameworks widen absorption while promoting dissociation [63], fluorenone acceptor cores pair efficient exciton dissociation with a well-defined active center [36], and even 3D COFs, long excluded by their sp^3 nodes, now sustain intrinsic metal-free photocatalysis [102]. Third, heteroatom decoration acts as a local-field lever: fluorination raises surface area, narrows the gap, and improves separation [54], with positional isomerism of fluorine alone moving AQY to 4.09% at 500 nm [143]; chlorine placement creates halogen-adjacent carbon HER sites active in seawater without cocatalyst [61]; and push–pull side groups correlate directly with ESIPT-mediated activity across matched six-membered substituent series [41, 45]. Local polarization engineering in truxenone frameworks [37], dipole modulation of vinylene COFs [98], and nitrogen-position tuning that reverses framework polarity [32] all act on the same underlying variable. The design rule: maximize the net (not nominal) D–A dipole per unit cell, verify with E_b or built-in field measurements, and only then attribute activity gains to the D–A lever [4, 81].

2.3 Linkage protonation and acid engineering

Protonation is the cheapest band-structure intervention available: exposing imine or azine lone pairs to Brønsted acids red-shifts absorption, stabilizes the excited state, and switches on exciton dissociation channels. The landmark demonstration lifted D–A imine COFs to 20.7 mmol g^{-1} h^{-1} upon linkage protonation [114]; HCl-protonated COF-920 subsequently reached 364 mmol g^{-1} h^{-1} under UV–visible light [17, 18]. The lever composes with others: simultaneous protonation and conjugation enhancement multiplies to an 88.4-fold gain over the parent framework [38], surface protonation plus interfacial engineering yields 18.23 mmol g^{-1} h^{-1} without any cocatalyst [20], and protonation raises conductivity and accelerates H_2 generation at Ru sites in vinylene frameworks [123]. The boundary condition is stability: protonation equilibria drift under operation, so durability claims require post-reaction structural verification (Section 7).

2.4 Defect and vacancy engineering

Beyond backbone substitution, deliberately installed defects create charge funnels: amide-functionalized defective CTFs reach 17.6 mmol g^{-1} h^{-1}, a 6.3-fold gain over the pristine framework, while preserving structural integrity [29], and missing-linker defects in porphyrin D–A COFs improve hydrogen evolution alongside oxygen reduction [117]. Crystallinity-resolved studies caution that defect and order effects are entangled: molecular engineering of tunable-crystallinity series shows activity tracking order parameters rather than nominal composition [108], which is why Section 4 treats crystallinity control as a first-class synthesis variable rather than a nuisance.

3 Mechanistic Characterization: Testing the Causal Chain

A design chain is only as credible as the measurements that can falsify each link. This section reviews the toolbox in the order the chain itself proposes – factor isolation first, then carrier dynamics, then atomic-scale and operando interface probes – and adopts calibrated language throughout: spectra *show* observables, global analysis *assigns* components under a model, and matched activity correlations *are consistent with* rather than prove causality [12].

3.1 Factor deconvolution before spectroscopy

Because COF photocatalysts co-vary crystallinity, porosity, absorption, wettability, and charge extraction, single-material reports rarely identify which factor moved. The structure–property–activity protocol of Ghosh and co-workers – quasi-independent variation of factors across the S-COF/FS-COF benchmark family – identified crystallinity and charge extraction as the prime HER determinants and remains the template for benchmarking claims [26, 101]. Substituent series on fixed backbones extend the same logic: six-membered substituent sets isolate electronic effects from framework geometry [45], and matched isoreticular linker series separate linker length from core identity [17]. Reviews of the crystallinity–stability dilemma emphasize that order parameters must be reported alongside rates for any comparison to be meaningful [85, 108].

3.2 Carrier dynamics: from femtoseconds to seconds

Time-resolved spectroscopy provides the quantitative observables of the middle links. Femtosecond transient absorption (fs-TA) resolves exciton formation, dissociation, and trapping; in S-scheme heterojunctions it directly tracks interfacial electron flow, as shown for ZnSe quantum dot/COF [137] and FS-COF/WO_3 hybrids [133]. Time-resolved photoluminescence, surface photovoltage, and wavelength-dependent photocurrent action spectra together assigned the enhanced activity of g-C_3N_4/Ni-CAT junctions to transfer of high-energy-level electrons rather than to absorption gains [13]. Nanoscale COF colloids show molecule-like excitonic behavior in photoluminescence and transient absorption, tying carrier physics to particle dimension [136], and lifetime engineering through D–A pairing is read out as dramatically prolonged carrier lifetimes accompanying the 82.6% AQE benchmark [51]. The historical caution from oxide photocatalysis applies unchanged: lifetime changes alone are not productive charge transfer unless they correlate with quantum yield under matched intensity [62, 94].

3.3 Atomic-scale and operando interface probes

Where the chain meets the cocatalyst, aberration-corrected HAADF-STEM and X-ray absorption fine structure establish speciation: atomically dispersed Pt on TpPa-1 was shown to bind as a six-coordinated C_3N–Pt–Cl_2 species, with DFT connecting that geometry to the activity gain [14]. Soft X-ray absorption identified halogen-adjacent carbons as the active H^* binding sites in Cl-decorated frameworks [61], and in situ XPS traced the S-scheme electron pathway in fully conjugated COF/C_3N_4 junctions [57]. Mechanistic assignment of the Volmer–Tafel route with a Tafel rate-determining step on Ru single-atom/nanocluster hybrids shows how kinetic analysis complements spectroscopy [84]. FT-IR and ^1H NMR located hydrogen-bonded nanoreactor encapsulation inside COF pores [75], while ESIPT-induced photoisomerization of keto-enamine linkages – invisible to steady-state diffraction – was caught by excited-state spectroscopy, reshaping how activity differences within the Tp family should be interpreted [105].

3.4 What counts as mechanistic evidence

Three practices from the strongest studies deserve adoption as field norms. First, matched controls: physical blends and amorphous analogues distinguish framework effects from composition effects [66, 110]. Second, multi-probe convergence: charge-separation claims supported jointly by fs-TA kinetics, photoelectrochemical response, and activity correlations [40, 89, 127] are categorically stronger than single-technique assignments. Third, protocol-anchored quantum yields: AQE at stated wavelengths – 6.7% at 420 nm for alkene linkages [69], 2.53% for sp^2c overall splitting [11], 9.39% under piezo-assisted operation [22] – remain the only observables portable across laboratories, a theme Section 5 develops quantitatively [139].

4 Synthesis Methodology and Crystallinity Control

Every claim in Sections 2 and 3 is gated by synthesis: crystallinity, stacking order, and morphology are set in the reactor, and they co-determine absorption, exciton transport, and site accessibility [82, 85]. This section organizes the methodology by decreasing thermodynamic control and increasing practicality.

4.1 Solvothermal standards and crystallization modulation

Most COF photocatalysts are still grown solvothermally – sealed-tube, 100–120 °C, days-long reactions whose solvent pair, catalyst, and water activity control the error-correction equilibrium [3, 106]. The Tp toolbox illustrates the maturity of this control: 1,3,5-triformylphloroglucinol reliably delivers crystalline, stable β-ketoenamine frameworks across many diamines, which is a major reason Tp-COFs dominate the composite literature [92, 97, 112]. For imine frameworks, cocrystal engineering of the monomer with p-toluenesulphonic acid regulates crystal structure formation and enables hydrogen-bond-directed growth [5], while base identity and loading tune triazine- sp^2c condensation [125]. Alkalinity screening in sp^2c photocathode synthesis shows the payoff quantitatively: optimal conditions (110 °C, 30 μL pyridine, 2 days) yield photocurrents 2.5× higher than suboptimally crystallized batches of the same composition [44] – a reminder that inter-laboratory rate disagreements often reduce to crystallinity differences [26, 108].

4.2 Fast, mild, and scalable alternatives

The scale-up path runs through routes that abandon the sealed tube. Mechanochemical grinding delivers β-ketoenamine COFs at room temperature with moderate crystallinity and retained chemical stability, and mechanochemically synthesized Tp-COFs now support systematic structure–property studies in photocatalytic HER [2]. Flow synthesis of TpPa-1 guided by green-chemistry principles modulates structure continuously and points toward manufacturing-compatible production [111]. Microwave heating compresses heterojunction growth to minutes [24, 25, 130], and nanoconfined synthesis at surfactant-stabilized oil–water emulsion interfaces reconciles high reaction rate with morphology control for imine 2D COFs [65]. Post-synthetic and irreversible-bond strategies – Katritzky-type skeletal editing among them – expand the accessible linkage space beyond dynamic covalent chemistry [76]. Room-temperature and template-free routes remain the least developed but most scale-relevant frontier [106].

4.3 Stacking stabilization and morphology engineering

Layer slippage and exfoliation degrade the columnar π-channels that carry charge; stabilizing the stack is therefore a synthesis objective in its own right. PEG-threading of pores locks the ordered layer arrangement and preserves HER activity [140]. Downsizing to colloidal nano-COFs multiplies mass-normalized rates – up to 392 mmol g^{-1} h^{-1} – by shortening exciton diffusion paths and exposing surface sites [136]; nanosheet engineering of β-ketoamine frameworks with pore-confined sub-nanometer Pt enables overall water splitting at 0.23% solar-to-hydrogen efficiency [118]. Morphology also mediates composite quality: in situ growth on oxide scaffolds produces chemically bonded, ultrathin junction layers [73, 90], and coordination-directed

Table 2: Representative single-phase COF photocatalysts. Rates in mmol g^{-1} h^{-1} unless noted; n.r. = not reported in the audited record.

System (linkage)	Design lever	Light / donor	Cocat.	Rate	AQY / note
Protonated D–A imine COF [114]	linkage protonation	visible / n.r.	Pt	20.7	landmark lever demo
COF-920 (protonated) [17]	isoreticular + protonation	UV–vis / n.r.	n.r.	364	gas-collection verified
β-Ketene–cyano D–A nanosheets [51]	D–A pair, exfoliation	450 nm / n.r.	Pt	n.r.	AQE 82.6%
Nano-COF colloids [136]	nanoscale downsizing	visible / n.r.	Pt	392	33.3 μmol h^{-1} absolute
DBC-sp^{2}c-COF [122]	sp^{2}c Kagome lattice	visible / n.r.	Pt	105.3	80.0 in natural seawater
COF-954 [139]	oligo side chains	visible / n.r.	5 wt% Pt	137.2	191.7 (seawater, UV–vis)
TAPPy-DBTDP-COF [55]	dual acceptors	visible / n.r.	Pt	12.7	top D–A tier
TpPaCl [61]	halogen active sites	visible / n.r.	none	16.3	11.5 in real seawater
TP-COF-F-3 [143]	F positional isomer	500 nm / n.r.	n.r.	n.r.	AQY 4.09%
COF–alkene [69]	linkage isolation	420 nm / n.r.	Pt	n.r.	AQE 6.7%
d-CTF(CONH$_2$) [29]	designed defects	visible / n.r.	n.r.	17.6	6.3× vs. CTF-1
Pt@TpBpy-NS [118]	nanosheet + pore-confined Pt	visible / none	sub-nm Pt	n.r.	OWS, STH 0.23%
sp^{2}c-COF (OWS) [11]	full conjugation	420 nm / none	n.r.	n.r.	AQE 2.53%, OWS

assembly of MOF-on-COF architectures yields intimate interfacial contact [64, 79]. Porphyrin-COF platforms carrying single-atom metals show that site placement can be encoded synthetically rather than post-hoc [9, 39].

The methodological bottom line mirrors the mechanistic one: report the order parameters (PXRD width, surface area, stacking signature) with the same prominence as the rate, because the synthesis variables are the hidden covariates of the entire literature [26, 82].

5 Performance Comparison Under Explicit Protocols

5.1 Why raw rates do not rank materials

Reported HER rates for COF systems span from below 1 mmol g^{-1} h^{-1} to nearly 400 mmol g^{-1} h^{-1}, but the protocol variables – irradiation spectrum and intensity, sacrificial donor, cocatalyst identity and loading, catalyst mass, and dispersion state – move rates by larger factors than most design levers [26, 136, 139]. Mass-normalized rates fall as loading rises; apparent quantum yield (AQY) at a stated wavelength and solar-to-hydrogen (STH) efficiency are the only cross-laboratory portable metrics [11, 59]. Pt-size studies underline the danger of single-variable narratives: across three Tp-series COFs the photodeposited Pt particle size varied with band gap, yet HER rates showed no correlation with that size [100]. Tables 2 and 3 therefore list protocol columns explicitly and mark unreported fields as n.r.; entries are exemplars, not a leaderboard.

Table 3: Representative COF heterojunction and composite systems. Abbreviations: ZIS = $ZnIn_2S_4$; BP = black phosphorus.

System	Junction type	Light / donor	Cocat.	Rate	Note
Pt/S@TpPa-1 [83]	Schottky, low work function	420 nm / n.r.	Pt	n.r.	AQE 80.7%
Py-HOF/COF [23]	organic–organic	visible / n.r.	n.r.	390.7	9.2× vs. COF
NiCo-NC@TP-BD [75]	pore nanoreactor	visible / n.r.	none	79.0	395× vs. bare COF
TiO_2 NFs@COF [25]	S-scheme	visible / n.r.	n.r.	58.3	microwave growth
Ru(II)PS@Py-COF [127]	molecular complex	visible / n.r.	n.r.	51.3	2.5× vs. COF
Diquat-ETM COF [66]	integrated e^- mediator	visible / n.r.	n.r.	34.6	beats physical mix
CdS/TFPT-DHTH-COF [131]	direct Z-scheme	visible / n.r.	none	15.8	75× vs. COF
CdS/TpPa-COF (1D) [21]	S-scheme, directional	visible / n.r.	n.r.	n.r.	TpPa platform
P-COF-1/CTF [110]	2D–2D π–π	visible / n.r.	n.r.	14.1	2.5× vs. CTF
JUC-704@ZIS [128]	cocatalyst-free junction	visible / n.r.	none	9.46	seawater operation
HKUST-1/TpPa-1 [64]	MOF-on-COF, II-scheme	visible / n.r.	n.r.	n.r.	coordination-bonded interface
BP/COF (2D–2D) [113]	metal-free p–n	visible / n.r.	none	n.r.	cocatalyst-free

5.2 The cocatalyst-dependence spectrum

Because Pt cost and scarcity bound any deployment scenario [84], the field is best read as a progression along a cocatalyst-dependence axis. At the noble-metal end, engineering focuses on atom economy: single-atom Pt anchored through defined C_3N–Pt–Cl_2 coordination on TpPa-1 [14], heteroatom-tuned Pt single-atom frameworks [95], sulfur-chelated high-loading single atoms that remain atom-efficient (11.4 mmol g^{-1} h^{-1} at 12.1 wt%) [7], and low-work-function Pt junctions whose $\Delta G_{H^*} = -0.13$ eV optimality yields AQE 80.7% at 420 nm [83]. Earth-abundant single atoms follow: Co on COFs [103], Fe–Co dual atoms on COF/C_3N_4 [74], zinc single atoms bridged by copper clusters [99], spinel cocatalysts with engineered orbital interactions [126], and DFT screens that rationalize single-atom–support pairing [72, 121]. Molecular and cluster catalysts occupy the middle ground: cyclometalated Pt in vinylene frameworks [46], azide-anchored cobaloximes on COF-42 that outlast their physisorbed analogues [27], enantioselectively immobilized chiral Co complexes (5.70 mmol g^{-1} h^{-1} noble-metal-free) [42], Cu(II) mediators on chiral β-ketoenamine hosts [104], and Ru sites on triazine sp^2c platforms extending to electrocatalytic HER [123, 135, 138]. The cocatalyst-free end is growing fastest: halogen-site TpPaCl [61], protonated CdS/COF architectures [20], benzothiazole metal-COFs [134], JUC-70x@ZIS junctions in seawater [128, 129], black-phosphorus p–n junctions [113], and C_{60}@TpPa charge-transfer dyads [124].

5.3 Stability reporting

Durability claims currently mix hour-scale runs, cycle counts, and post-mortem characterization. The stronger reports pair retention percentages with structural verification: 12-hour photocurrent stability

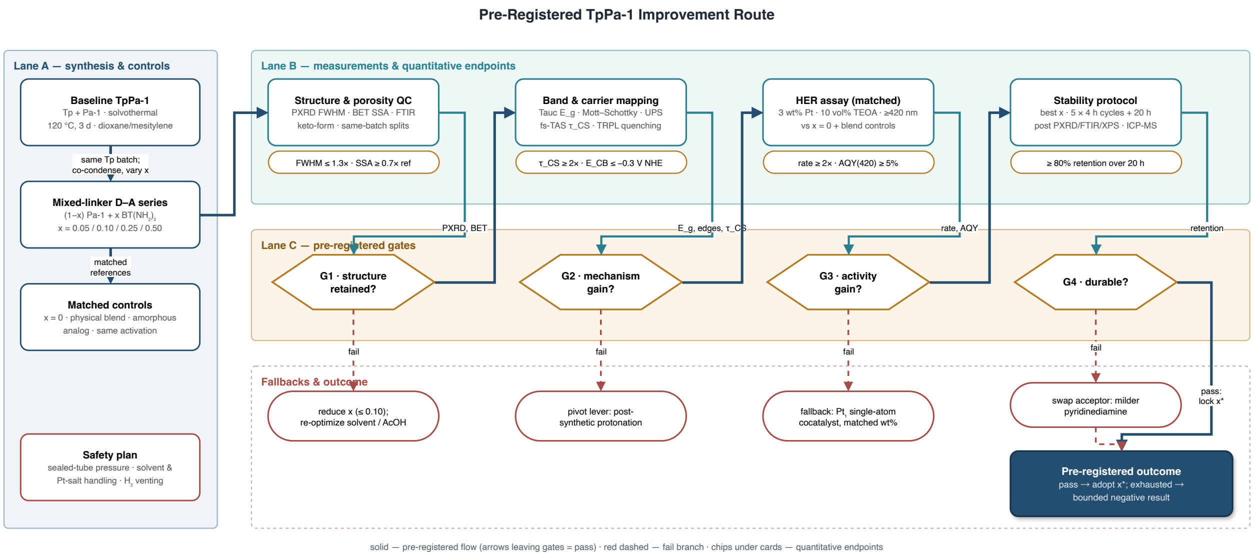

Figure 2: Pre-registered TpPa-1 improvement route. Lane A fixes synthesis and matched controls; Lane B specifies quantitative measurement endpoints; Lane C gates progression (G1–G4). Red dashed branches are pre-committed fallbacks; the outcome is adopted only if all gates pass, otherwise the result is reported as a bounded negative.

with retained crystallinity [58], sustained performance of integrated-interface frameworks [40], S-scheme junctions surviving repeated cycling [15, 16, 57, 88, 132, 141], and functional-group- resolved junction dynamics [31, 109]. Chirality-induced spin selectivity [107], piezo-assisted charge steering [22], and interfacial covalent bonding [50, 79, 119] each claim durability side benefits that still await standardized, long-duration verification – the protocol gap Section 7 addresses [19, 35, 93].

6 A Pre-Registered Improvement Route for TpPa-1

The design rules of Sections 2–5 are only useful if they generate falsifiable predictions. This section formulates one: a mixed-linker donor–acceptor doping route for TpPa-1, chosen because the baseline is the field's best-characterized β-ketoenamine platform [92, 97, 112] – stable in boiling water and strong acid, synthesizable solvothermally, mechanochemically, and in flow [2, 111] – yet its interrupted in-plane conjugation and modest exciton dissociation leave clear headroom that composites routinely exploit [16, 21, 64, 83, 124, 141].

6.1 Hypothesis and route

The causal chain predicts that installing a fraction x of an electron-accepting diamine – benzothiadiazole-diamine, $BT(NH_2)_2$ – into the Pa-1 position creates local D–A dipoles that lower the exciton binding energy and raise charge-separation yield, without abandoning the tautomer-locked stability of the Tp node (Figure 2, Lane A). The acceptor choice has direct precedent: benzothiadiazole-dianiline frameworks are established HER platforms whose activity responds sharply to local electronic modulation [10, 55]. Statistical copolymerization at $x \in \{0.05, 0.10, 0.25, 0.50\}$ under fixed solvothermal conditions (120 °C, 3 d, dioxane/mesitylene) holds the linkage chemistry constant while sweeping the dipole density – the quasi-independent factor-variation design that benchmark studies established [17, 26, 45]. Matched controls are pre-committed: the $x = 0$ baseline, a physical blend, and an amorphous analogue with identical composition and activation [66, 110].

6.2 Quantitative gates

Lane B fixes the endpoints before synthesis begins, and Lane C converts them into pass/fail gates. G1 (structure retained): PXRD line width within 1.3× of baseline and BET surface area at least 0.7× baseline, so that crystallinity – the dominant hidden covariate [44, 108] – cannot masquerade as an electronic effect. G2 (mechanism gain): charge-separation lifetime $\tau_{CS} \geq 2\times$ baseline by fs-TA with TRPL-quenching corroboration, and conduction-band edge at or below −0.3 V vs. NHE by Tauc/Mott–Schottky/UPS so the HER driving force is preserved [12, 13, 81]. G3 (activity gain): HER rate at least 2× baseline and AQY(420 nm) ≥ 5% under a matched assay (3 wt% Pt, 10 vol% TEOA, ≥420 nm) benchmarked against the $x = 0$ and blend controls [69, 100]. G4 (durability): ≥80% rate retention over 20 h in 5 × 4 h cycles with post-run PXRD/FTIR/XPS/ICP-MS verification, the stability standard the composite literature is converging toward [40, 58].

6.3 Fallbacks and interpretation

Each gate failure triggers a pre-committed pivot rather than post-hoc tuning (Figure 2, bottom): G1 failure reduces x and re-optimizes solvent acidity; G2 failure pivots the lever to post-synthetic protonation, which acts on the same link of the chain with independent precedent [20, 38, 114]; G3 failure swaps in an atom-economic single-atom cocatalyst at matched loading [7, 14]; G4 failure substitutes the milder pyridine-diamine acceptor. If all gates pass, the adopted x^* becomes the new baseline; if the fallback tree is exhausted, the result is reported as a bounded negative – the D–A dipole density is insufficient at preserved crystallinity for this family – which is itself a publishable constraint on the causal chain [26]. Safety is pre-registered alongside science: sealed-tube pressure, solvent handling, Pt-salt disposal, and H_2 venting follow the protocols embedded in the route plan.

7 Challenges on the Path to Scalable Solar Fuels

Following the style of the strongest outlooks, we phrase each challenge as a decision question with a measurement and a failure criterion rather than as a wish list [19, 93].

Can the field converge on portable metrics? The comparison problem of Section 5 is structural: rates without wavelength-resolved AQY, donor identity, and normalization basis cannot be aggregated [100, 139]. The decision question is whether journals and reviews require AQY(λ) and STH – with light-intensity linearity checks inherited from oxide photocatalysis [94] – as submission defaults. Failure looks like another decade of leaderboards built on incommensurable mmol g^{-1} h^{-1} values [1, 28].

Does activity survive the donor? Nearly all Table 2 entries consume sacrificial electron donors. Overall water splitting on COF platforms exists – sp^2c frameworks at AQE 2.53% [11], pore-confined Pt nanosheets at STH 0.23% [118], bifunctional Pt/COF systems evolving stoichiometric H_2/O_2 [130], and computationally paired HER/OER COF couples [96] – but efficiencies trail donor-assisted numbers by two orders of magnitude. The measurable milestone is STH ≥ 1% on an all-organic or COF-dominant system with verified stoichiometry; persistent reliance on TEOA-class donors within techno-economic models is the failure mode [6, 43].

Is long-duration stability real? Hour-scale retention with post-mortem PXRD/XPS is becoming standard [40, 58], but solar-fuel service means thousands of hours under flow, oxygen crossover, and protonation drift (Section 2.3). The decision experiment is a 1000-hour continuous run with periodic structural audit – the COF analogue of module aging tests; crystallinity–stability trade-offs [82, 85] and linkage hydrolysis under operando acidity are the leading failure hypotheses.

Can synthesis leave the sealed tube? Gram-scale, days-long solvothermal batches cannot supply reactor arrays. Flow synthesis of TpPa-1 [111], mechanochemical batches [2], microwave composite growth [24], and emulsion-interface morphology control [65] define the candidate manufacturing set; the milestone is kilogram-scale production retaining Section 4 order parameters at <10% penalty [49, 106].

Can noble metals be engineered out? The Pt-dependence spectrum of Section 5 is trending in the right direction – single-atom economy [14, 95], earth- abundant single atoms [74, 103, 121], molecular Co catalysts [27, 42], and genuinely cocatalyst-free platforms [61, 113, 128, 134] – but matched-protocol parity with 3 wt% Pt baselines remains the unambiguous target [84].

Do photons reach the framework in a reactor? Photoelectrode and panel formats change the photon economics that powder assays hide: COF photocathodes sustain tens of μA cm^{-2} [44, 58], oxide-scaffolded junctions manage light and charge jointly [35, 73], and PEC-oriented reviews now treat 2D COFs as absorber layers in their own right [30, 77]. Reactor-level milestones – areal STH under natural irradiance, seawater tolerance [122, 128, 139] – should replace mass-normalized records as the field's headline numbers [72, 142].

8 Conclusion

Organizing a decade of COF photocatalysis along a single causal chain – linkage → conjugation → band structure → exciton and charge separation → interface → HER – turns a sprawling literature into a set of testable design rules. Linkage chemistry no longer forces a stability–conjugation trade-off: vinylene and sp^2-carbon frameworks deliver both, while β-ketoenamine and imine platforms remain the workhorses for systematic study [11, 47, 112]. Donor–acceptor dipole engineering, verified by exciton-binding measurements rather than nominal composition, and linkage protonation are the most portable band-structure levers [51, 81, 114]. Mechanistic claims increasingly rest on multi-probe convergence with matched controls [13, 26], and the strongest performance reports anchor rates to AQY(λ) with explicit protocol variables [83, 136]. What stands between laboratory quantum efficiency and solar fuels is no longer molecular designability but verification infrastructure: portable metrics, donor-free operation, thousand-hour stability, scalable synthesis, and reactor-level photon management [93, 106]. The pre-registered TpPa-1 route of Section 6 is offered as a template for how the field can convert its accumulated design rules into gated, falsifiable experiments – and how negative results, bounded by matched controls, can constrain the causal chain as productively as records [69, 100].

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